Online Odyssey Outlet - Applying Systems Thinking for Metric Assessment

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Problem Statement

You are a junior analyst working for an e-commerce startup, Online Odyssey Outlet.

The date is January 2020, and the startup has just been through its first winter sale. Senior stakeholders are interested in knowing which products performed best during the Christmas period so that they can streamline the products they advertise in future sale periods.

You will do this by analyzing up to two months of events data, presented as two separate data sources with identical structures. Events refer not just to sales but to more general customer actions, such as viewing a product on the website or placing it into their virtual cart.

The challenge is that, when pressed, stakeholders aren't exactly sure what they mean by "best." In an initial brainstorming session, they highlight some aspects they care about when ranking their products:

- Volume of sales
- Total revenue from a single product
- Popularity, measured by the number of unique customers who bought a product
- Conversion, meaning the percentage of time a product is bought once it has been placed in the virtual shopping cart
- Products with increased performance from November to December

Systems Thinking

1. Understanding the problem :

Senior stakeholders are interested in knowing which products performed best during the Christmas period so that they can streamline the products they advertise in future sale periods.

What are our hoping stakeholders to achieve if they advertise the best performing products :

- To attract and retain new customers
- To improve sales
- To increase revenue
- To increase brand awareness and presence
- e.t.c

What system are we breaking down ?

- System to attract and retain new customes
- System to improve sales
- System to improve revenue
- System to increase brand awareness/presence

Elements of the system ?

- Customers
- Products
- Store Workers
- e.t.c

Activities by customers in the online store sales system ?

1. Browse items :

What else could happen as customers browse items ?

- Website can crash if there is too much traffic
- They see products that got no stocks left
- They see product prices
- They see products that are available and with stock
- They compare product prices
- They read descriptions
- They read reviews
- They check product ratings
- They click and view individual products and read descriptions .
- Compare product specs .
- They add products to cart
- They remove products from carts
- They buy/checkout the products in carts

2. What else could happen when customers do the above activities :

Webcrash :

- Potential sales get lost if there is a web crashes .
- Pressure increases for the web team to fix the site .
- Web quality can get re-evaluated / web team can get reassessed / Employees can lose their jobs .

- Customers lose faith / trust / loyalty in our brand .
- We lose customers / potential repeat sales .
- Bad word of mouth spreads if there is a web crash .
- Customers get discouraged to revisit the site .
- Customers may think the whole business got shut down .

Stock-Outs:

- They get frustrated / complain / leave / look for product elsewhere when they see the product they looking for with no stock .
- They lose faith/loyalty in our business when they see products with no store .
- Business lose potential sales from stockouts .
- Inventory management get reassessed . Restocking pressure can increase .

Prices :

- They buy product that are a little cheaper .
- High value products lose potential sales .
- High value products can get removed from the store catalogue .
- Read reviews and descriptions to see if the products are worth the price . They can then get encouraged/discouraged to buy the item .

Reviews/Description Reading :

- They decide whether to buy or not , decide whether to trust/test the product judging by its reviews and ratings .
- Products with no reviews / no descriptions can not be taken into consideration/ discourage customers from buying .

Add to Cart :

- Products in cart get reviewed before check-out .
- Can remove products in cart , if the total amount is above their budget or change their mind on a product .
- They can leave without checking out .
- They can reset the whole cart .
- They can proceed to check-out .

Check-Outs :

- They can stop the whole check-out process if they not comfortable with given out their personal infomation .
- They can make mistakes in when typing details such as name , delivery addresses e.t.c. Which leads to delivery delays . Delays lead to bad WOM , reviews , complaints , lose of trust , lose of future customers .
- They can request refunds . Lose potential revenue . Receive a bad complaint / review .
- They wait for their orders to get delivers .
- They can reschedule their deliveries .

- They complain if deliveries dont arrive on promised date .
- They can cancel their orders .
- Check-outs increases number of orders/deliveries store employees have to handle . Stress levels increase , then packaging mistakes and damages . Customers receive incorrect items , which leads to returns / complaint / bad WOM . Also decreases product inventory , which leads to stock-outs .

Product Arrival :

- Delivery personal can get lost if not given proper addresses / can damage the items , e.t.c .
- They can test products , return product if they not what they expected .
- They can recommend people to our store .
- They can share experience with others .
- They can write reviews on the products .

Activities by store employees :

1. Handle orders/deliveries :

What could happen as store workers handle/deal with orders :

- Product inventory drops . Product stock-outs occur
- Wrong items can be delivered to wrong customers .
- Store workers can get stressed as workload increases . Products can get damaged . Employees can get warnings / lose their jobs , Damages get deducted from employee wages . People can resign as they get overworked or burnout . Conflict between employees can occur . Scandal occurs which turnish brand reputation .

Stocks from activities :

For each activity , what can be measured ? What depletes or accumulate ? What else can be infered for the activity's stock ?

1. Browsing :

Number of Customers browsing (Web traffic) Number of new customers Number of repeat customers Product impression Web traffic capacity

2. Web Crash :

Number of web crashes Web quality / server quality Web traffic capacity Customer trust Brand reputation Lost Sales Brand awareness / presence Web team stress levels Web team workload Web team quality Web team skill Web team experience Job loss Scandal / Conflicts Brand reputation Customer churn Bad word of mouth

3. Stock-Outs :

Product stock-outs Product Inventory Purchase Intent Lost sale Customer Trust Customer disatisfaction Customer complaints

4. Price :

Product Price Purchase Intent Customer trust Complaints Brand reputation Customer satisfaction Product Cost Product Profit

5. Descriptions :

Product description length Product description quality/clearity Purchase Intent Customer Trust Product quality (from specs)

6. Reviews :

Product rating Number of good reviews Number of bad reviews Purchase Intent Customer Trust Brand reputation Customer satisfaction

7. Click/Views

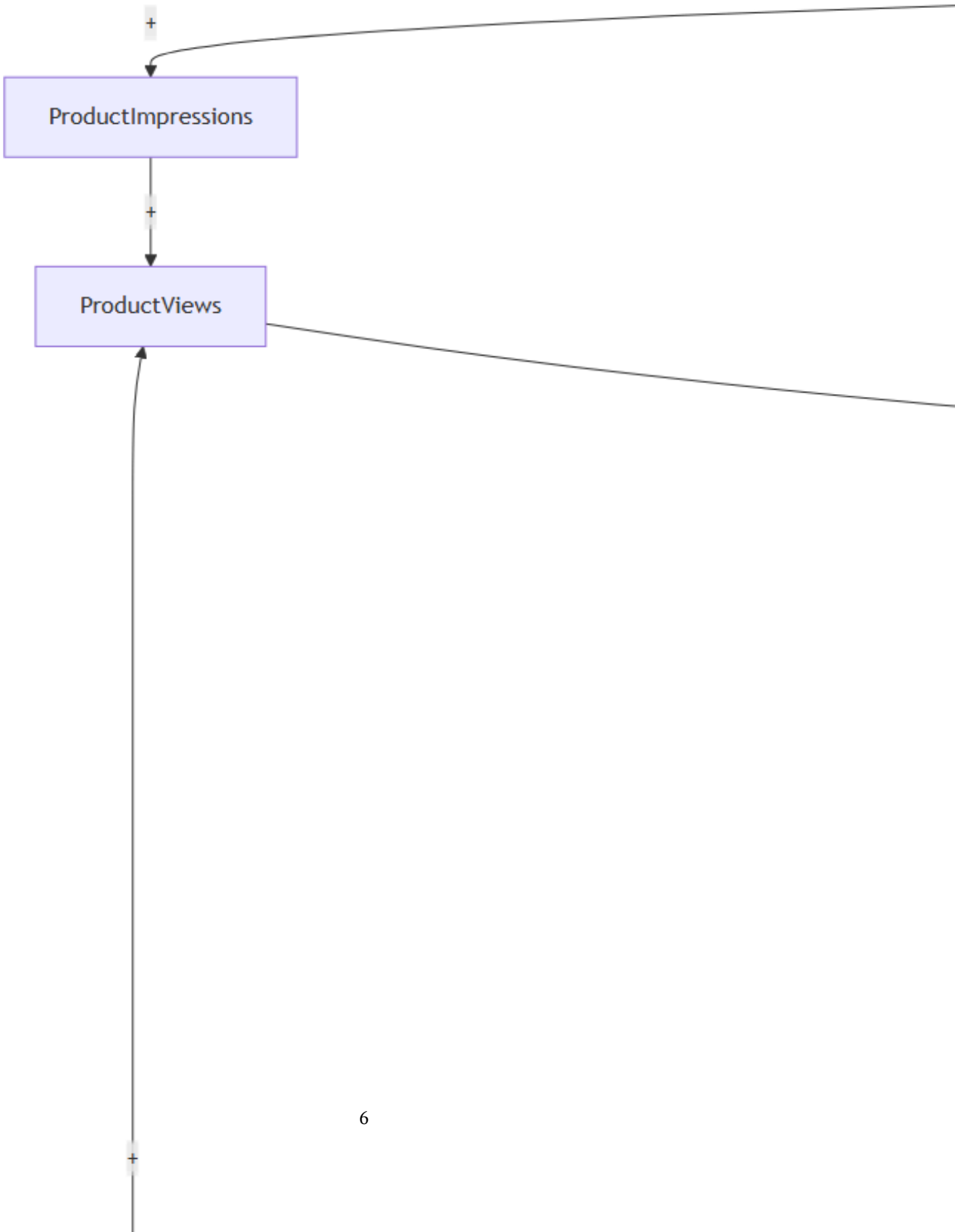
Product clicks Products views Product appeal

8. Cart Add / Removal

Number of times product got added to cart Number of times product got removed from cart Purchase Intent Customer Budget Cart total Product appeal

9. Check-Outs

Number of store orders Employee workload Employee stress level Product inventory Number of store deliveries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ProductImpressions
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